

# Trace Tables

Project: Transparent Book Recommender

## 1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Branching Logic

Without opening the old lesson, explain one key idea from Branching Logic. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Precision in Algorithms

Without opening the old lesson, explain one key idea from Precision in Algorithms. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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## 2. Practise with fading help

Use W2: Trace before running for Loops and trace tables. Record the answer once; the lesson and workbook refer to the same attempt.

**Given inputs · Procedure I chose · Middle steps · Result and check**

**Given inputs:** \_\_\_\_\_

**Procedure I chose:** \_\_\_\_\_

**Middle steps:** \_\_\_\_\_

**Result and check:** \_\_\_\_\_

\_\_\_\_\_

**The step I completed without help, and the hint I needed**

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### 3. Individual reasoning check

Use the worked case for Loops and trace tables. Proposed explanation: “The stop value is included”.

Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

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### 4. Independent transfer case

**Independent case: Build a repeated addition independently**

A box starts with 5 counters. Each round adds 2 counters. Use rounds = 3, an initialized total and for turn in range(rounds). Print the final total once after the loop. Trace the totals and predict fresh runs with rounds 0, 1 and 3. Assume rounds is a nonnegative integer.

**My independent explanation, trace and limits**

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

## 5. Your learning target

Start score=1; add 4; if score>3 subtract 2; double score. Record score after every operation. A student evaluates the condition using the initial value. Locate the divergence.

## 6. Solve the given case

Set score=2; add 3; if score>4 subtract 1; double score.

Trace every value.

Show your trace, calculation or evidence

**Write or draw your reasoning and evidence.**

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## 7. Check a changed case

Set score to 2. Add 3, then double the result. Record the score after each operation. What changes if you double before adding?

My independent answer and reason

**Write or draw your reasoning and evidence.**

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**Week 4 Test and reflect**

**8. Design a revealing test**

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

**Test record · My evidence**

**Test record:** Changed input or condition

**My evidence:** \_\_\_\_\_

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**Test record:** Expected answer or fallback

**My evidence:** \_\_\_\_\_

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**Test record:** Observed result

**My evidence:** \_\_\_\_\_

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**Test record:** Why the result occurred

**My evidence:** \_\_\_\_\_

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**9. Project checkpoint**

Trace each book's genre, length, difficulty, and total score before sorting.

The project change and evidence I will keep

**Write or draw your reasoning and evidence.**

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## 10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review \_\_\_\_\_ Next step \_\_\_\_\_

## 11. Transfer practice from the original programme

Start score=1; add 4; if score>3 subtract 2; double score. Record score after every operation. A student evaluates the condition using the initial value. Locate the divergence.

### My independently reasoned answer

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Record support used: none, hint or teacher explanation.

## 12. W1 Explain the range

List the values supplied by range(2, 5) and explain why 5 is absent. How many times does its body run?

### My reasoning and evidence

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## 13. W2 Trace before running

Copy the worked script. Change the range to range(2, 5). Record turn, total before and total after each iteration. Predict the single output, then run and compare.

### My reasoning and evidence

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## 14. W3 Repair the reset

For the faulty script with range(1, 4), record total after each addition. Explain the wrong final output and remove only the faulty reset.

### My reasoning and evidence

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## 15. W4 Build a repeated addition independently

A box starts with 5 counters. Each round adds 2 counters. Use rounds = 3, an initialized total and for turn in range(rounds). Print the final total once after the loop. Trace the totals and predict fresh runs with rounds 0, 1 and 3. Assume rounds is a nonnegative integer.

### My reasoning and evidence

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## 16. W5 Check a contract limitation

In your counter script, `range(-2)` performs no iterations. Does returning the initial 5 make negative rounds valid? Explain what the script checks and what it assumes.

### My reasoning and evidence

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## 17. Feedback, return check and learning record

### Evidence · My record

**Evidence:** Worked alone / hint / model

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** First step I repaired and why

**My record:** \_\_\_\_\_

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**Evidence:** New case checked later; date and result

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** What I can now explain without notes

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** One remaining question

**My record:** \_\_\_\_\_

\_\_\_\_\_

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

## 18. Pictorial case practice: Trace Tables

A fictional reading-token jar starts with one token and gains three per round.

Python: `total=1; for turn in range(4): total=total+3; after the loop print(total).`

The update is inside the loop; print is outside.

## Loop totals after each iteration

|        | turn | total after update |
|--------|------|--------------------|
| First  | 0    | 4                  |
| Second | 1    | 7                  |
| Third  | 2    | 10                 |
| Fourth | 3    | 13                 |

Read a cell using both its row and column labels.

## Trace Tables

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Initial state

Before the loop, total=1.



#### 2. Iteration values

range(4) supplies turn values 0,1,2,3.



#### 3. Updated totals

After each turn total is 4,7,10,13.



#### 4. After the loop

Print 13 once, after all four updates.

### CHANGE ONE THING

Use range(0) instead of range(4).

Explain the changed result using the matching lesson.

Move print inside the original loop. What sequence is displayed?

### My picture-based explanation and changed-case prediction

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Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Trace recommender loops with current item, eligibility, score and current best so each update can be explained.